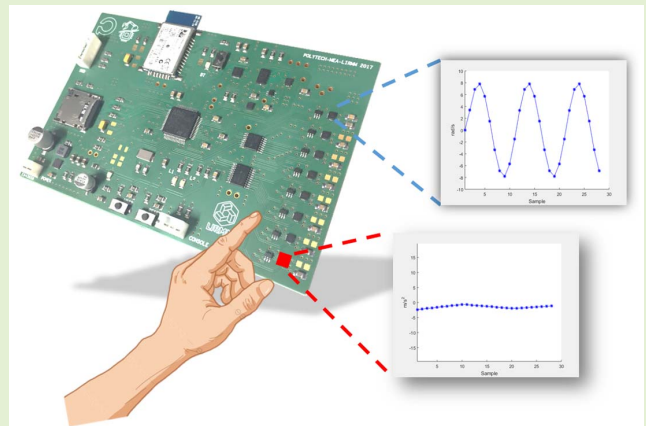


A New Algorithm for Fault Tolerance in Redundant Sensor Systems Based on Real-Time Variance Estimation

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Abstract—Low cost and small size integrated sensors bring a significant interest to redundant sensing systems such as sensor array systems. Redundancy may permit to increase both performance and dependability of individual sensors at the system level, however, the emergence of these redundant systems has generated the need for dynamic algorithms able to detect and act against the presence of errors in system elements. Then, a new adaptive algorithm for sensor array systems is presented in this work. This approach is generic and can be implemented for different types of sensor systems, besides it does not require any information about the inputs. The proposed real-time algorithm is based on the following: 1) estimation of sensor individual variances at each time step, 2) identification and removal of sensors affected by errors, and 3) reincorporation of sensors after recovery or replacement. Variance estimation is revisiting Minimum Norm Quadratic Unbiased Estimation (MINQUE) method to allow real-time operation and requires that the number of sensors is strictly greater than two times the number of signals to be measured. Consequently, the proposed method is able to support the presence of $m-1$ faulty devices, where m is the total number of sensors less two times the number of measured signals. Performance of the proposed algorithm is demonstrated through theoretical demonstrations, simulations, and experimental results.

Index Terms—Variance estimation, sensor array system, homogeneous systems, MINQUE, data fusion, fault detection, weighted average, Kalman filter.



I. INTRODUCTION

A SENSOR array system is defined as a collection of same-type (usually called as homogeneous system) or different-type sensors (called heterogeneous system), usually deployed in identical or complementary pattern and used to measure one or several physical inputs [1]. Once measurements are obtained, a data fusion method is applied to leverage redundancy in measurements. These methods allow to exploit the benefits of redundant information, such as reaching better performance than that achieved individually by each sensor

and adapting to their possible failures / uncalibrations [2]. These two characteristics are essential for the development of autonomous sensing systems.

Array signal processing and measurement data fusion have been extensively studied in the literature. For example, in [3], a fusion method for measurements from several triads of accelerometers and gyroscopes is presented. Simulations assuming arrays of four accelerometer / gyroscope triads are done, and an embedded system of 32 Micro-Electro-Mechanical Systems (MEMS) based accelerometer / gyroscope / magnetometer triads is used to show experimentally that redundancy in data can overcome low-cost sensor problems. This algorithm exploits the advantage of redundancy in data obtained from different types of inertial sensors, keeping this solution restricted to inertial applications.

A more generic solution is proposed in [4], where an algorithm based on random weighting is presented. Here, a weighted average based on the analysis of variances of individual sensors is implemented for data fusion. A similar approach is presented in [5], where the weight factors problem is solved by using trained neural networks. These approaches are interesting in terms of genericity since they can be applied

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to any redundant sensor array system, nevertheless, they are unreliable under sensors failures.

This point is particularly important since besides smart array techniques can enhance the performance offered by using individual sensors [6], it is known that such techniques could potentially enable in-run fault detection; overcoming the fact that, with the exception of very few high-end devices, most of the electronic sensors do not integrate reliable self-diagnostic capabilities [7]. This crucial issue constitutes a research aspect where it is intended for sensors to reach a level they could operate for substantially long operation times in the absence of updates and/or corrections, which would be a leap.

An example of this is found in [7] where physical redundancy is implemented by mounting two counter-aligned accelerometers to perform a differential measurements. Here, redundancy in measurements is used to reduce the bias thermal sensitivity, as well as to define a simple fault detection method by comparing the difference between the signals from the two accelerometers to a threshold.

Other methods, such as the characterization of errors for their subsequent identification, have been exploited. For example, in [8] a method to characterize the noise of all MEMS accelerometers in an array system by generating controlled excitations is presented. Similarly, in [9] a method to detect, identify, and quantify faults is proposed. Such method implements a likelihood ratio test to identify faulty sensors. Once identified, the fault is classified by modeling it and then comparing it against a predefined list of error models. This proposal makes use of the parity space approach, where the idea is to take advantage of the fact that there are more sensors than signals to measure.

From the state of the art, it was observed that weighted average presents two main interests: i) the estimation of sensors variances without the knowledge of the input and, ii) the genericity of the algorithm. In addition, it was observed that the use of the parity space approach permits the detection of sensor faults. By studying previous work in those fields [4], [9], we propose: 1) to improve the estimation process of individual variances of sensors avoiding the use of a golden device, and 2) to introduce reliability constraints into the algorithm to handle dynamically the presence of faulty sensors.

As a result, a new real-time algorithm has been developed for fault tolerance and noise reduction in redundant sensor arrays. This algorithm is revisiting Minimum Norm Quadratic Unbiased Estimation (MINQUE) [10] for real-time to bring several advantages with respect to state-of-the-art:

- Unlike MINQUE, estimation of sensors variances can be done at each time step, permitting to track the effects of low-frequency noise,
- As in MINQUE, knowledge of the input signal is not required,
- Unlike in [4], the use of a golden device for the variance estimation is not required,
- Genericity, i.e., it can be implemented for different types of sensor arrays,

- On-line detection of systematic errors is carried out under certain circumstances,
- Unlike [5], fault tolerance is implemented as long as the number of fault-free sensors is greater than two times the number of measured signals.

This paper is organized as follows: In section II, the problem statement is presented. Then, the proposed algorithm for estimating variances in sensor array systems is detailed in section III. In section IV, fault detection and reincorporation of recovered devices is explained. Finally, in V and VI the assessment of the proposed algorithm by means of simulations and a real implementation is presented.

II. PROBLEM STATEMENT

Consider a discrete-time sensor array system S composed of n same-type sensors, all of them measuring a linear combination of the same r physical quantities denoted as X (n and $r \in \mathbb{N}_1$). Then, measurement reported by sensor $i \in S$ at time step k is defined as [1]:

$$y_{i,k} = (\alpha_T + \alpha_{i,k})h_i X_k + \beta_{i,k} + \varepsilon_{i,k} \quad (1)$$

where:

- α_T is the typical value for the scale factor (usually normalized to one)
- $\alpha_{i,k}$ is the scale factor error of sensor i at time step k
- h_i is the observation row vector of size $1 \times r$
- X_k is the $r \times 1$ vector of physical quantities to measure
- $\beta_{i,k}$ is the bias of sensor i at time step k
- $\varepsilon_{i,k}$ denotes the measurement noises of sensor i (white noise, flicker noise, brown noise, etc.)

Observe that $\alpha_{i,k}$ and $\beta_{i,k}$ may change over the lifetime of an implementation. Hereafter, the term *uncalibration* refers to values of $\alpha_{i,k}$ and / or $\beta_{i,k}$ different from zero. On the other hand, observation vector h_i is different than the scale factor $(\alpha_T + \alpha_{i,k})$; h_i corresponds to the intensity at which sensor i receives inputs (usually linked to the topology of the array), while $(\alpha_T + \alpha_{i,k})$ is given by the physical properties of the sensor. Vector h_i is usually known, while the value of $\alpha_{i,k}$ can only be determined by a calibration of sensor i . From here on, it will be assumed that h_i is known.

To simplify the analysis, assume that all sensors in S are initially calibrated. This means that scale factor errors and bias errors are zero. Then, measurement reported by calibrated sensor i at time step k can be reformulated as:

$$y_{i,k} = h_i X_k + \varepsilon_{i,k} \quad (2)$$

Assuming that all sensors in S are synchronized, the vector of n measurements at time step k is defined as:

$$Y_k = H X_k + E_k \quad (3)$$

where: $Y_k = [y_{1,k}, \dots, y_{n,k}]^T$, $H = [h_1, \dots, h_n]^T$, and $E_k = [\varepsilon_{1,k}, \dots, \varepsilon_{n,k}]^T$. From here, two cases can be listed: 1) measurement of a single input (physical signal), and 2) measurement of two or more inputs. To illustrate the transcendence of estimating variances in the data fusion process

carried out by an array of sensors, examples of the two cases listed above are presented next.

A. Sensor Array Measuring a Single Input

When $r = 1$, X denotes a scalar whose best estimate is obtained by computing a weighted average of all sensors [11]. Assuming that stochastic errors of different sensors are independent and not related with the input, a weighted average of all sensors at time step k is described by:

$$\hat{X}_k = w_{1,k}y_{1,k} + \dots + w_{n,k}y_{n,k} \quad (4)$$

where $w_{i,k}$ denotes the weight of sensor i at time step k . Besides, weights are subject to the constraint $\sum_{i=1}^n w_{i,k} = 1$.

For simplicity, assume that X is present in all sensors with the same intensity, i.e. $H = 1_n$, where 1_n denotes all-ones vector of size n . Then, using (2), equation (4) can be re-expressed as follows:

$$\hat{X}_k = X_k + \sum_{i=1}^n w_{i,k} \varepsilon_{i,k} \quad (5)$$

Then, variance of the estimation error is:

$$\sigma^2(\text{error}_k) = \sum_{i=1}^n w_{i,k}^2 \sigma^2(\varepsilon_{i,k}) \quad (6)$$

where $\sigma^2(\varepsilon_{i,k})$ denotes variance of sensor i at time step k . Finally, estimation of X_k is obtained by minimizing (6). Such minimization results in a relation between weights and variances of all sensors in the system, as presented below [4], [11], [12]:

$$w_{i,k} = \left(\sum_{j=1}^n \frac{\sigma^2(\varepsilon_{i,k})}{\sigma^2(\varepsilon_{j,k})} \right)^{-1} \quad (7)$$

Equation (7) represents the well known weighted average method, where all weights are computed at each time step by means of the variances of all sensors. Once weights are computed, input signal is estimated using (4).

B. Sensor Array Measuring Multiple Inputs

When $r \geq 2$ using a weighted average becomes complex and sometimes even impossible. However, inside the literature exist several methods for multiple signal estimation, from where, one of the most popular is the Kalman filter [13].

Kalman filters use a predictor-corrector type estimator that address the general problem of estimating vector X by combining vector of measurements described in (3) and a prediction of X_k given by [14]:

$$X_k^- = \Phi \hat{X}_{k-1} + v_{k-1} \quad (8)$$

where Φ is the $r \times r$ state transition matrix which relates X_{k-1} with X_k , and v_{k-1} is the r size process noise vector. Estimation of X_k is carried out by relating the measurement described in (3) and prediction given in (8):

$$\hat{X}_k = X_k^- + K_k(Y_k - HX_k^-) \quad (9)$$

where:

$$\begin{aligned} K_k &= \mathcal{P}_k^- H^T (H \mathcal{P}_k^- H^T + R)^{-1} \\ \mathcal{P}_k^- &= \Phi \mathcal{P}_{k-1} \Phi^T + Q \\ \mathcal{P}_{k-1} &= \mathcal{P}_{k-1}^- - K_{k-1} H \mathcal{P}_{k-1}^- \\ Q &= \text{diag}(\sigma^2(X)) \\ R &= \text{diag}([\sigma^2(\varepsilon_{1,k}), \dots, \sigma^2(\varepsilon_{n,k})]) \end{aligned} \quad (10)$$

Here, $\text{diag}(v)$ denotes a square diagonal matrix with the elements of vector v on the main diagonal. In [15], it is shown that accurate estimates of Q and R are crucial to obtain a good estimate of X . Moreover, if the implementation has a poor approximation of Φ , or if a nonparametric method (such as the one presented in [16]) is used, then the knowledge of R becomes decisive for a good performance of the filter.

III. ESTIMATION OF VARIANCES IN A SENSOR ARRAY

In this section, we propose a modification of MINQUE method (presented for the first time in [10] and more recently extended in [17]) to allow real time operation. Our approach leverages correlation in the collection of sensors in a system that meets the three following constraints: 1) all sensors must be initially calibrated, 2) all devices must measure the same physical inputs, and 3) the number of sensors must be two times greater than the number of measured inputs. This method is detailed below.

Consider array system S described in section II. It is assumed that the number of sensors in S is greater than the number of inputs to measure (i.e. $n > r$). Then, measurement vector Y_k is described by (3) where it is assumed that H is known. Thus, due to the redundancy, Y_k lies in a subspace of dimension r . Let P be the orthogonal projection matrix onto the orthogonal complement of H (the method to obtain this matrix is described later in subsection III-A). Hence, P is a $n \times n$ matrix with rank $n - r$ that satisfies the constraint: $PH = 0$. Thus, the multiplication of P with Y_k results as:

$$PY_k = PE_k \quad (11)$$

Let $\sigma^2(v_k)$ be an on-line function that computes the variance of each element in a vector v from the first time step to the k -th time step, i.e. $\sigma^2(v_k) = \frac{1}{k} \sum_{j=1}^k (v_j - \bar{v})^2$, where $\bar{v} = \frac{1}{k} \sum_{j=1}^k v_j$. If this function is applied to (11), then the resulting variance is:

$$\sigma^2(PY_k) = P^{\circ 2} \sigma^2(E_k) \quad (12)$$

where $P^{\circ 2}$ denotes the Hadamard product of P with itself (i.e., $P^{\circ 2} = P \circ P$). The result is a $n \times n$ matrix with full rank (in section III-B it is specified for which cases $P^{\circ 2}$ is non-singular), and therefore, invertible. Consequently, variances of sensors at time step k can be estimated as follows:

$$\hat{\sigma}^2(E_k) = (P^{\circ 2})^{-1} \sigma^2(PY_k) \quad (13)$$

Expression above is the main result of this work: a real time method based on MINQUE. Computation of P at each time step is not required unless the topology of the array changes.

A. Orthogonal Projection P

The definition of matrix P depends entirely on the definition of H , which at the same time, depends entirely on the physical characteristics of the system, such as the topology, resolution of the sensors, etc. These characteristics are usually known, and therefore, computation of P can be performed straightforwardly. Considering that H is the $n \times r$ known matrix with rank equal to r , computation of P is carried out as follows [18]:

$$P = I_n - H(H^T H)^{-1} H^T \quad (14)$$

where I_n is the identity matrix of size n . Again, computation of P is not required unless addition or removing measurements from the data fusion process coming from a specific sensor is needed.

B. Existence of $(P^{\circ 2})^{-1}$

To show that the Hadamard product $P \circ P$ is invertible, first, three properties of P are highlighted:

Lemma 1: The orthogonal projection matrix P is symmetric and idempotent.

Lemma 2: A symmetric and idempotent matrix is positive semi-definite with eigenvalues of 0 and/or 1.

Theorem 1 (Schur product theorem): Suppose P is a square positive semi-definite matrix of size n . Then, $P \circ P$ is also positive semi-definite.

Although theorem 1, $P^{\circ 2}$ is still invertible under certain conditions. For example, it is known that for $n \leq 2$, the Hadamard power $P^{\circ 2}$ cannot be positive definite [10], but the situation is different when $n \geq 3$. In [19], some interesting results related to Hadamard products are presented, from which the next sufficient condition for the non-singularity of $P^{\circ 2}$ is taken:

Lemma 3: Let P be a positive semi-definite matrix of size $n \times n$, with $n \geq 3$. If P has no zero main diagonal entries, then $P \circ P$ is positive definite if there is a $q \in \{1, \dots, n\}$ such that $\text{rank}(P) > n - q$ and every principal minor of P of size q is positive.

Here, to ensure that every principal minor (hereinafter denoted as PM) of P with order q is positive is equivalent to guarantee the existence of a constant q such that, every list of q distinct columns of P is linearly independent [19]. Now, the proof of an interesting property that arises from the previous lemma is presented:

Lemma 4: To ensure the non-singularity of $P^{\circ 2}$, it is required that $2r < n$.

Proof: Due to the fact that $\text{rank}(P) = n - r$, and $\text{rank}(P) > n - q$, then $r < q$. In addition, it should be noted that $q \leq n - r$ (otherwise, the list of q distinct columns from P contains dependent columns), this implies that $2r < n$.

Significantly, this result shows that to carry out the estimation of variances by means of a based-MINQUE method, the number of sensors used for this process needs to be strictly greater than twice the number of measured signals. Besides, note that $r < q \leq n - r$. This means that, assuming that P has no zero diagonal entries, and that $n > 2r$, to check the existence of $(P^{\circ 2})^{-1}$ it is enough to check whether every PM of size $q = r + 1$ is positive.

Summarizing, for array systems composed of 2 sensors it is not possible to implement the MINQUE method [10]. For systems composed of 3 or more sensors, to ensure that $P^{\circ 2}$ is non-singular, it is sufficient to verify the next three conditions: 1) the number of sensors is greater than two times the number of measured signals, 2) each element in the main diagonal of P is different from zero, and 3) all the PM of P of order $q = r + 1$ are positives. Finally, we highlight the following particular case:

Lemma 5: Assuming $n \geq 3$, if $r = 1$ then $P^{\circ 2}$ is non-singular.

Proof: For the sake of simplicity, assume that all sensors perceive the input X with the same intensity, meaning that $H = 1_n$. The proof comes from the fact that, due to the definition of H , P can be defined in terms of n as follows [20]:

$$P = \begin{cases} \frac{n-1}{n} & , \text{ for } P_{i,j} \mid i = j \\ \frac{-(n-1)}{n(n-1)} & , \text{ for } P_{i,j} \mid i \neq j \end{cases} \quad (15)$$

Observe that all elements in the main diagonal of P are different from zero, fulfilling the second constraint. Then, the non-singularity of $P^{\circ 2}$ is ensured by verifying that all PM of size $q = 2$ are positives. This is guaranteed by the fact that $\left(\frac{n-1}{n}\right)^2 > \left(\frac{-(n-1)}{n(n-1)}\right)^2$.

Thus, for an array system composed of three or more sensors used to measure a single physical input, this method can be implemented without the need of verifying the non-singularity of $P^{\circ 2}$.

IV. DETECTION OF SYSTEMATIC ERRORS

Assuming an initial calibration, errors in measurements are mainly due to stochastic phenomena; however, failures and uncalibration of sensors may appear in runtime. The estimation of individual variances carried out with (13) is sensitive to the presence of systematic errors, which allows their detection under certain scenarios. To show this, first let's rewrite equation (3) as follows:

$$Y_k = \text{diag}(A_k) H X_k + B_k + E_k \quad (16)$$

where $A_k = [(a_T + \alpha_{1,k}), \dots, (a_T + \alpha_{n,k})]^T$, and $B_k = [\beta_{1,k}, \dots, \beta_{n,k}]^T$. Because of the initial calibration, $\alpha_{i,k}$ and $\beta_{i,k}$ are equal to zero $\forall i \leq n$ and $k < \mathcal{K}$, where $\mathcal{K}$ denotes the time step at which the first systematic error is present. Thus, when measurement vector Y_k is multiplied by the orthogonal projection P , the following is obtained:

$$\begin{cases} P Y_k = P \text{diag}(\alpha_k) H X_k + P B_k + P E_k & \text{if } k \geq \mathcal{K} \\ P Y_k = P E_k & \text{if } k < \mathcal{K} \end{cases} \quad (17)$$

where $\alpha_k = [\alpha_{1,k}, \dots, \alpha_{n,k}]^T$. Equation 17 and the assumption about an initial calibration show that errors due to the presence of biases ($B_k \neq 0$) are detected directly due to their equivalence to a low-frequency noise variation (i.e., $P B_k \equiv P E_k$). Now, to analyze the case of scale factor errors, equation (17) is simplified for $k \geq \mathcal{K}$:

$$P Y_k = P \text{diag}(\alpha_k) H X_k + P E_k \quad (18)$$

Thus, when one or more sensors are affected by scale factor errors, the variance of PY_k results in:

$$\sigma^2(PY_k) = (P \text{diag}(\alpha_k) H)^{\circ 2} \sigma^2(X_k) + P^{\circ 2} \sigma^2(E_k) \quad (19)$$

To guarantee the correctness of the algorithm, it is required to make the assumption that at each time step there is at most one uncalibrated sensor. Following this restriction, the expression above is simplified to:

$$\sigma^2(PY_k) = P^{\circ 2} \text{diag}(\alpha_k^2) H^{\circ 2} \sigma^2(X_k) + P^{\circ 2} \sigma^2(E_k) \quad (20)$$

Then, the estimation of sensors variances given by $(P^{\circ 2})^{-1} \sigma^2(PY_k)$ results as follows:

$$\hat{\sigma}^2(E_k) = \text{diag}(\alpha_k^2) H^{\circ 2} \sigma^2(X_k) + \sigma^2(E_k) \quad (21)$$

Thus, if there is a single device affected by scale factor error then the estimation of sensors variances remains undisturbed, except for the one corresponding to the uncalibrated sensor, i.e.:

$$\begin{cases} \hat{\sigma}^2(E_{i,k}) = \alpha_k^2 h_i^{\circ 2} \sigma^2(X_k) + \sigma^2(\varepsilon_{i,k}) & \text{uncalibrated} \\ \hat{\sigma}^2(E_{i,k}) = \sigma^2(\varepsilon_{i,k}) & \text{otherwise} \end{cases} \quad (22)$$

Based on this, the use of a threshold on the estimated variances for the discrimination of uncalibrated devices is proposed:

$$\begin{cases} \hat{\sigma}^2(E_{i,k}) \leq \text{threshold} \Rightarrow \text{calibrated sensor} \\ \hat{\sigma}^2(E_{i,k}) > \text{threshold} \Rightarrow \text{mark as uncalibrated} \end{cases} \quad (23)$$

Once marked, a sensor is removed from the variance estimation process for the following steps. This is done by removing its measurement from Y and its corresponding row from H , as well as recomputing P from the new definition of H .

Equations 22 and 23 show that the ease of detecting the uncalibrated sensor is linked to the magnitudes of the input signals variances. For example, suppose that a threshold $= 3\sigma_j^2$ (with σ_j^2 denoting the variance corresponding to the noisiest sensor in the system) is setup. Then, assuming a single measured signal and that all sensors perceive it with the same intensity (as in section V), the uncalibrated device (denoted by i) will be detected only when $\frac{3\sigma_j^2 - \sigma^2(\varepsilon_{i,k})}{\alpha_i^2} < \sigma^2(X)$.

Finally, to summarize: assuming that i) $P^{\circ 2}$ is invertible, and that ii) for any time step $k \geq \mathcal{K}$ there is at most one unmarked uncalibrated sensor, then this method is able to give a correct estimation of variances corresponding to the faulty-free sensors, detect bias errors (and the devices linked to these), as well as identify sensors affected by scale factor errors.

A. Reincorporation of Sensors

Once an uncalibrated sensor is found, it is removed from Y_k and H , and therefore, its variance is no longer computed. However, to detect recovered sensors (sensors that have been replaced or re-calibrated at runtime) variances of uncalibrated sensors are required. Such variances can still be computed

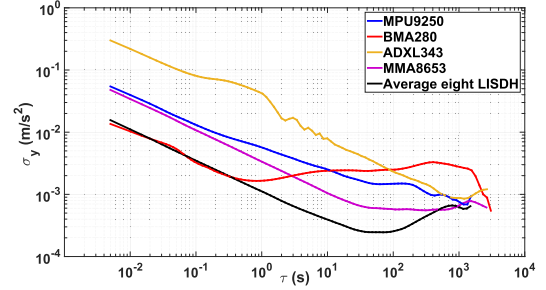


Fig. 1. Allan deviation graph corresponding to the five different models of sensors.

using (13), however, it must be guarantee the presence of at most one uncalibrated sensor in Y_k to ensure the correctness of this method. This is done as follows:

Let $\tilde{S}_{k-1} \subset S$ be the set of all sensors that remain marked as uncalibrated at time step $k-1$. For each sensor $j \in \tilde{S}_{k-1}$ at time step k :

- Reincorporate measurement reported by sensor j at time step k to Y_k
- Reincorporate the corresponding observation row of sensor j to H
- Compute matrix P and check if $P^{\circ 2}$ is invertible
- If $P^{\circ 2}$ is invertible then compute $\hat{\sigma}^2(E_{k,j})$ using (13), otherwise sensor j cannot be added
- Once estimated, the variance of sensor j is used to determine if there is a recovery or not by evaluating expression (23): if sensor j is still detected as uncalibrated then it is marked again and removed from Y_k and H

V. EXPERIMENTAL VALIDATIONS

The proposed algorithm is implemented in an in-house embedded system composed of twelve MEMS-based chipsets: one MPU9250, one BMA280, one ADXL343, one MMA8653, and eight LIS2DH (see Figure 2). Each chipset is a three-axis accelerometer with options to different parameterizations. For these experiments, a full scale of $\pm 2g$ and an output data rate of 200Hz are set up for all the devices. Characterization of the stochastic phenomena present in the five different models of accelerometers is carried out using an Allan DEViation (ADEV) graph [1] (Figure 1). These curves are obtained using the AVAR Matlab function [21] and sets of 10^6 samples coming from sensors outputs, setting a zero input and a sampling time of 5 ms.

Assuming an initial calibration and that sensors axes are aligned with the axes of the array frame, the standard model equation for the accelerometers can be expressed as follows [3], [22]:

$$a_{i,k} = a_{\text{body},k} + \dot{\omega}_k \times d_i + \omega_k \times (\omega_k \times d_i) + \varepsilon_{i,k} \quad (24)$$

where $a_{i,k}$ is the acceleration measured by the i -th triad at time step k ; $a_{\text{body},k} = [a_{\bar{x},k}, a_{\bar{y},k}, a_{\bar{z},k}]^T$ is the acceleration of the body frame in axes $\bar{x}$, $\bar{y}$ and $\bar{z}$; ω_k and $\dot{\omega}_k$ are the angular velocity and angular acceleration of the body frame; d_i is the position of i -th triad in the array coordinate frame; and $\varepsilon_{i,k}$ is the vector of noise present in the triad. For

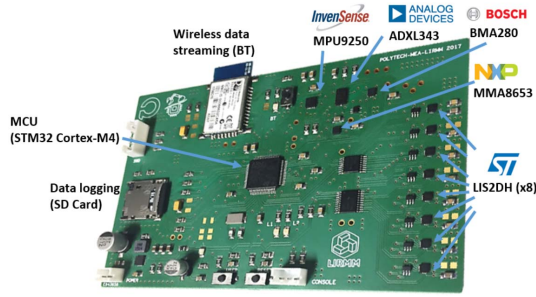


Fig. 2. In-house designed embedded system with an inertial sensor array composed of four sensor chipsets. Each chipset contains an accelerometer triad.

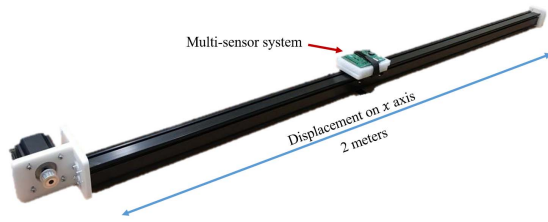


Fig. 3. Rail built for acceleration measurement using the card presented in Figure 2

the experiments, a rail that allows a linear displacement in only 2 directions (forward and backward) was designed (see Figure 3). By aligning the array coordinate frame with the rail, the sensor array system is exposed to accelerations only along one of the axes (hereinafter the $\vec{x}$ axis). This makes $a_{\vec{y},k}$, $a_{\vec{z},k}$, ω (and therefore, $\dot{\omega}$) equal to zero. Thus, the measurement reported by i -th sensor at time step k is simplified as follows:

$$a_{i,k} = a_{\vec{x},k} + \varepsilon_{i,k} \quad (25)$$

Because it presented the lowest noise level during characterization (Figure 1), it was decided to use the average of the eight LIS2DH sensors outputs as the reference to estimate the errors in the rest of the sensors. Therefore, the proposed algorithm is implemented using the following parameters: $Y_k = [a_{\text{MPU9250},k}, \dots, a_{\text{MMA8653},k}]^T$, $n = 4$, $X = a_{\vec{x},k}$ (and therefore $r = 1$), matrix $H = 1_n$, and $m < 2$. Due to $r = 1$, the inverse of $P^{\circ 2}$ exists as long as $n \geq 3$ (lemma 5). Thus, the implementation of the proposed algorithm for this scenario supports only one faulty/uncalibrated sensor. For the input, a random signal with amplitude $< 2g$ is manually generated. The length of the experiment is 1000 s.

For estimating the variance described in Equation 12, an on-line method that computes the variance of each element in a vector, assuming that vector elements are independent samples that belong to different discrete sequences, is used [23, p. 232]. Such method is implemented with an exponential smoothing instead of an incremental counter, where a smoothing factor $\gamma = 0.999$ is empirically selected.

The assessment of the proposed method is addressed to three main scenarios: 1) fault-free sensors, 2) system impact due to biases, and 3) system impact due to scale factor errors. The first scenario is evaluated directly using outputs from the

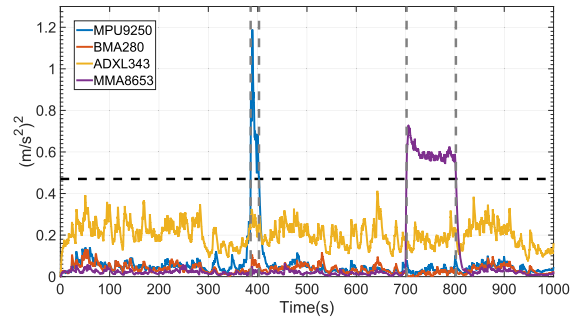


Fig. 4. Individual variance estimation of a system composed of four accelerometers. Vertical dotted lines show uncalibration/recovery identifications. Horizontal dotted line show the threshold for such identification.

sensors. However, for scenarios 2 and 3, systematic errors in the sensor outputs have been induced by software.

Thus, the length of the experiment is divided into three sections: 1) from $0s \leq t \leq 300s$ it is assumed that all sensors are free of systematic errors, 2) a scale factor error $\alpha_{\text{MPU9250}} = -0.2$ is induced in the MPU9250 sensor between 300s and 400s, then a re-calibration is assumed; and 2) a bias $\beta_{\text{MMA8653}} = 1.5 \text{ m s}^{-2}$ is induced in the MMA8653 sensor between 700s and 800s, then a re-calibration is assumed.

The first scenario is based on the assumption that only stochastic phenomena affect sensors measurements. Figure 4 presents the sensors variances estimations obtained with the proposed algorithm. Such estimations correspond to the noise levels shown in Figure 1. For uncalibrated sensors detection (scenarios 2 and 3), threshold described in (23) is arbitrarily set as: $\zeta = 2 \sigma^2(\varepsilon_{\max})$, where $\sigma^2(\varepsilon_{\max})$ denotes the average of the highest variance in S (variance of ADXL343) estimated within the interval $0s \leq t \leq 100s$. After 100s, once the variance of any sensor exceeds this threshold, the algorithm assumes the presence of a new uncalibrated sensor and the corresponding procedure for removing measurements coming from such sensor is carried out. In the same way, if the value of any uncalibrated sensor falls below the threshold, then the algorithm assumes a re-calibration and its measurements are reincorporated to the measurement vector. In Figure 4, uncalibrations on sensors are detected using the threshold $\zeta = 0.4702(\text{m/s}^2)^2$ (horizontal dotted line). Vertical dotted lines in gray represent the detection of the uncalibration in sensor MPU9250 at 385s, its recovery at 402s, the detection of a fault on sensor BMA280 at 702s, and its recovery at 802s. Observe that, for this example, the scale factor error detection took longer than the bias error detection. This is because the detection of scale factor error is strongly related to the variance level of the input signal.

Finally, the advantage of error detection using estimated variances is demonstrated through an example of estimation of the input signal. For this, the proposed algorithm is used together with the weighted average method presented in (7). This method is compared against the estimation obtained through a simple average (using Y_k) and the Least Squares (LSQ) method. For the LSQ, data obtained from Y_k and the reference given by the eight sensors LIS2DH between

TABLE I
COMPARISON OF DIFFERENT METHODS FOR THE INPUT ESTIMATION

	RMS error in m/s ²		
	Fault-free	Scale factor error	Bias error
	0s to 300s	300s to 400s	700s to 800s
Simple average	0.3164	0.2624	0.4494
LSQ method	0.3016	0.2601	0.7295
Weighted average	0.3018	0.2318	0.2924

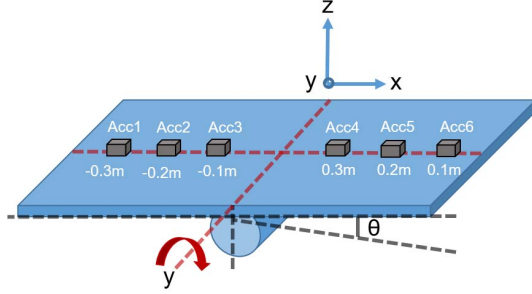


Fig. 5. Tilt sensing using six single-axis accelerometers.

the interval $0s \leq t \leq 100s$ is used to calculate the coefficients. For the assessment, the Root Mean Square (RMS) error is used as the performance indicator. Table I shows the RMS errors obtained using the three methods mentioned above. From this table, for the fault-free sensors scenario, the performance of the proposed method is better than the one obtained by the simple average, and it reaches the one obtained with the LSQ method, in real-time and without input knowledge (unlike the LSQ method that requires input knowledge). Once failures occur, the estimate obtained using the proposed algorithm shows a better performance than the other two methods.

VI. SIMULATION OF A MULTIPLE INPUTS SCENARIO

The previous section illustrates the usefulness of the proposed algorithm as a data fusion method when the sensing system is used to measure a single input. When two or more input signals are estimated, weighting average using the proposed algorithm is not suitable, however, estimation of sensors' variances still can be done, enabling the proposed method to be implemented together with other data fusion algorithms, such as the Kalman filter.

To exemplify this, an analysis of the proposed algorithm's implementation for a sensor array system used to estimate two input signals is presented. For this analysis, simulations inside the Matlab Simulink environment are used. The problem to handle for this analysis is tilt and angular velocity sensing, previously presented in [24], [25], and briefly described below.

Consider a horizontal platform with a fulcrum at the center capable of changing its tilt angle by rotating about the y-axis (see Figure 5). For such rotation, let's assume a sinusoidal angle variation with amplitude ± 1 rad and a frequency of 1 rad s^{-1} .

Given this scenario, tilt angle and angular rate have to be estimated at each time step. Due to their nature, gyroscopes are by default the sensors used for tilt sensing. However, gyroscopes present high angle estimation errors due to angular

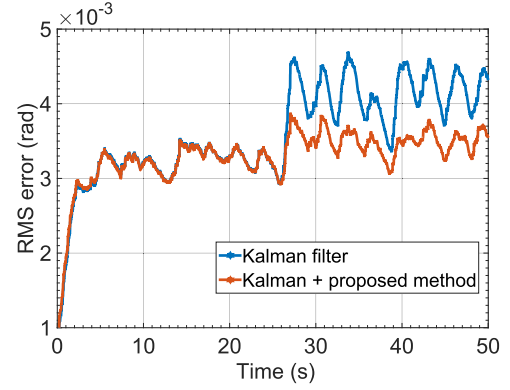


Fig. 6. Error in θ estimation. In blue Kalman filter parameterized with static variances, in red the same filter parameterized with dynamic variances computed with the proposed algorithm.

velocity data bias accumulation over time [24]. To overcome this, gyroscopes are often replaced by accelerometers to measure the tilt of an object in a static environment. Contrary to gyroscopes, accelerometers data have lower long term drift.

Based on this, a discrete-time system composed of six x-axis accelerometers is used for the estimation of the following two parameters: 1) the tilt angle of the platform (denoted by θ), and 2) the angular rate (denoted by ω). Using the center of the platform as reference, sensors are placed at: -0.3 m , -0.2 m , -0.1 m , 0.1 m , 0.2 m , and 0.3 m (see Figure 5). Consequently, each accelerometer measures both the tilt angle and the angular rate as follows:

$$\text{Acc}_{i,k} = g \sin(\theta) + \omega_k^2 d_i \quad (26)$$

where g is the gravity vector, $\text{Acc}_{i,k}$ denotes the measurement reported by accelerometer i at time-step k , and d_i is the algebraic distance between the i -th sensor and the platform center. From Equation 26 it is unable to determine the sign of ω . This could be solved by including a gyroscope in the measurement system and the Kalman filter. However, for this simple example, we consider the estimation of ω as sufficient, regardless of its rotation sense.

For the estimation of θ and ω , a Kalman filter is used. From the described above, and based on equations (8) to (10), the following configuration for the Kalman filter is obtained:

$$X_k^- = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sin(\theta) \\ \omega^2 \end{bmatrix}_{k-1}$$

$$Y_k = \begin{bmatrix} \text{Acc}_{1,k} \\ \vdots \\ \text{Acc}_{6,k} \end{bmatrix} = \begin{bmatrix} g & d_1 \\ \vdots & \vdots \\ g & d_6 \end{bmatrix} \begin{bmatrix} \sin(\theta) \\ \omega^2 \end{bmatrix}_k$$

Then, θ and ω are computed from the estimated $\sin(\theta)$ and ω^2 , remembering that for this example the sign of ω is omitted. The transition noise matrix is setup as $Q = \text{diag}([0.001, 0.001])$ during simulations. Finally, for the measurement noise matrix, two different configurations are assessed:

- Static variances predefined after multiple tests (0.0001 for accelerometers),

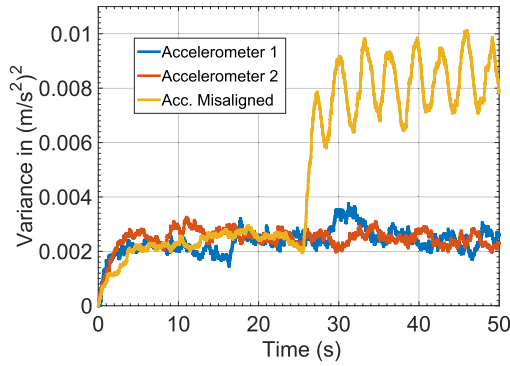


Fig. 7. Example of the runtime estimation of the accelerometers' variances using the proposed algorithm.

- Dynamic variances computed at runtime by the proposed algorithm.

On the other hand, based on the scenario description, the parameterization of the proposed method is: the number of sensors $n = 6$, where measurement vector is $Y_k = [\text{Acc}_{1,k}, \dots, \text{Acc}_{6,k}]^T$, the number of input signals $r = 2$, where $X_k = [\sin(\theta), \omega^2]_k^T$. H results in a $n \times r$ matrix defined as $H = [9.81, -0.3; \dots; 9.81, 0.3]$.

As mentioned before, simulations are carried out inside Matlab Simulink environment. For sensors simulation, the model presented in [24] is used, which is capable of simulating different types of noise (white, pink and brown noises) and systematic errors. Then, for the six accelerometers, the following model's parameterization is used: Noise = $0.005 \text{ m s}^{-2} \text{ Hz}^{-1/2}$, Bias instability = 0.002 m s^{-2} , rate random walk = $7 \times 10^{-5} \text{ m s}^{-2} \text{ Hz}^{1/2}$, bandwidth = 300 Hz, and sampling frequency = 100 Hz. The way to parameterize sensor models is explained in [24].

Finally, to emphasize the usefulness of the proposed algorithm, we assume a misalignment error in one of the sensors occurred during execution time. For this, a 10° clockwise z-axis rotation (denoted by ψ) is introduced in one of the sensors, causing the following change in the accelerometer x-axis output:

$$\text{Acc}x_{i,k} = g \sin(\theta) \cos(\psi) + \omega_k^2 d_i \cos(\psi) \quad (27)$$

Now that the scenario has been presented, analysis of the simulation results is performed. Figure 6 shows the RMS error in θ estimation for a simulation time of 50 s. In this figure, two magnitudes are presented: 1) error in blue corresponds to the Kalman filter parametrized with static variances, and 2) error in red corresponds to the Kalman filter parametrized with dynamic variances computed with the proposed algorithm. Figure 6 shows that:

- For free-faulty cases, the estimation error is the same or even lower when variances computed with the proposed algorithm are used inside the Kalman filter.
- The misalignment error (induced after 25 seconds of the simulation) is suppressed when the Kalman filter is used in combination with the proposed algorithm. This does not occur when static variances are used

TABLE II
SIMULATION RESULTS

Kalman filter	RMS error	
	Estimation of θ	
	Before misalignment	After misalignment
Using static variances	0.0033 rad	0.0042 rad (+27%)
Using variances computed with the proposed method	0.0033 rad	0.0035 rad (+6%)
	Estimation of ω	
	Before misalignment	After misalignment
Using static variances	0.2019 rad/s	0.2226 rad/s (+10%)
Using variances computed with the proposed method	0.1952 rad/s	0.1962 rad/s (+1%)

instead. To illustrate this, Figure 7 presents the estimated variances (by means of the proposed algorithm) of three of the accelerometers, where one of them is the accelerometer affected by the alignment error. Note that the algorithm is sensitive to the presence of such misalignment, detecting an increase in the variance of the involved sensor and without disturbing the other estimates.

Table II summarizes the results obtained during simulations in terms of the RMS error. Such table present RMS errors concerning the estimation of both θ and ω , using the Kalman filter with both, static and dynamic variances. Observe that the use of the proposed algorithm not only improves the accuracy of the Kalman filter, but also gives it robustness against the presence of runtime errors.

VII. CONCLUSION

In this paper, a new method designed for sensor arrays is proposed. Such method is used to classify sensing elements according to their noise level, as well as to detect and handle faults/recoveries during runtime. This algorithm can be implemented together with a weighted average or a Kalman filter for carrying out a data fusion process. To implement this method, three conditions must be met: i) the number of sensors must be greater than three, ii) the number of sensors must be greater than two times the number of measured signals, and iii) matrix $P^{\circ 2}$ (the hadamard power of two of the orthogonal projection of the observation matrix) must be invertible. The first two conditions can be straightforward met, however, the third condition requires more attention and here the conditions to verify the non-singularity of such matrix are presented. In addition, when a sensor array is used to measure a single input, $P^{\circ 2}$ was proven to be always invertible. On the other hand, in order to give robustness to the algorithm, a more efficient way of discriminating uncalibrated sensors must be found, as well as seeking to inhibit the restriction of handling one uncalibrated sensor at a time step. These improvements are intended as future extensions of this work.

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